

```
FROM node:14
```

```
WORKDIR /app
```

```
COPY package.json ./
```

```
RUN npm install
```

```
COPY . .
```

```
EXPOSE 3000
```

```
CMD ["npm", "start"]
```

Create the GitHub Actions workflow: In your repository, create a new file `.github/workflows/docker-ci-cd.yml` and define the workflow as follows:

```
# .github/workflows/docker-ci-cd.yml
name: CI/CD Pipeline for Docker
```

```
on:
  push:
    branches:
      - main
```

```
jobs:
  build:
    runs-on: ubuntu-latest

    steps:
      - name: Checkout code
        uses: actions/checkout@v2

      - name: Set up Docker Buildx
        uses: docker/setup-buildx-action@v1

      - name: Log in to Docker Hub
        uses: docker/login-action@v2
        with:
          username: ${ secrets.DOCKER_HUB_USERNAME }
          password: ${ secrets.DOCKER_HUB_PASSWORD }

      - name: Build Docker Image
        run: |
          docker build -t myusername/myapp:latest .

      - name: Run Tests
        run: |
          docker run --rm myusername/myapp:latest npm test

      - name: Push Docker Image
        run: |
          docker push myusername/myapp:latest
```

```
run: |
  docker push myusername/myapp:latest
```

Let's explain this. The workflow is triggered when there are pushes to the `main` branch. In response, the `build` job runs on the latest version of Ubuntu.

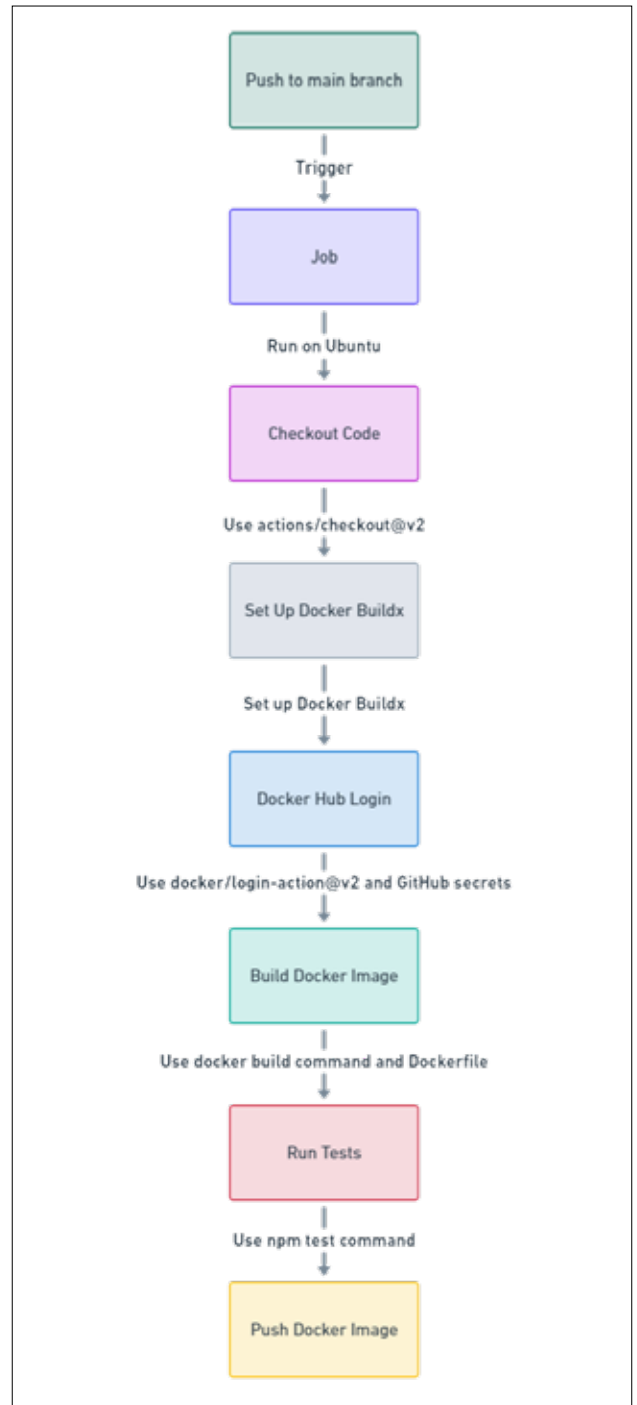


Figure 1: Explanation of the workflow